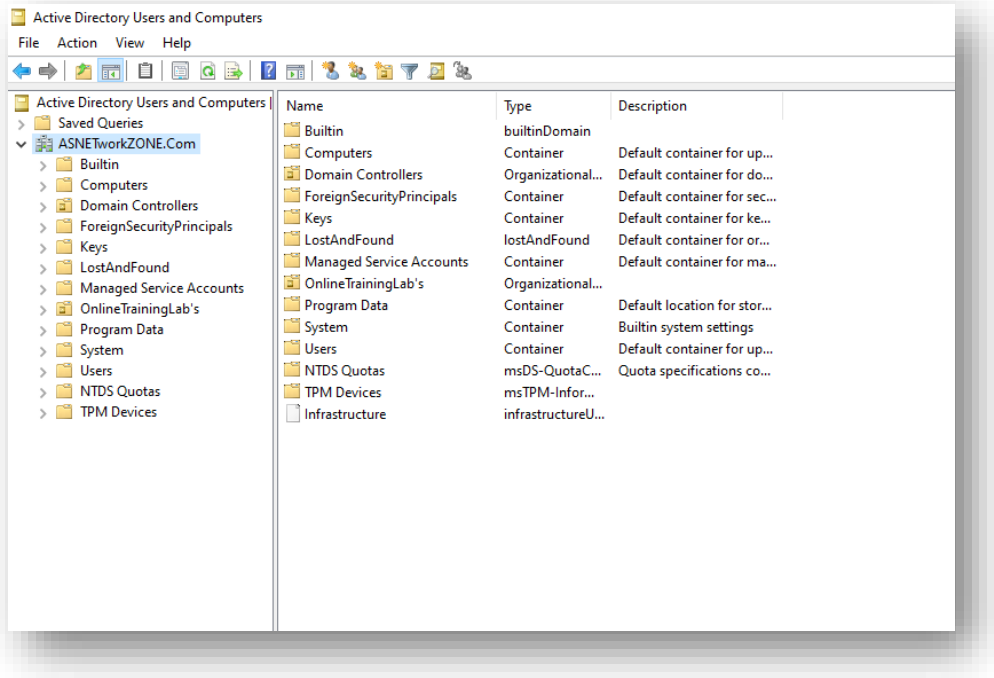
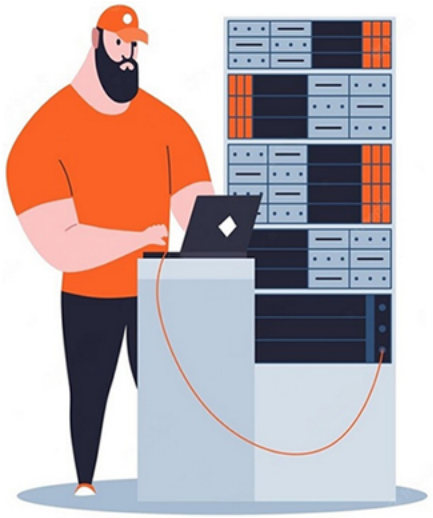
AGENDA

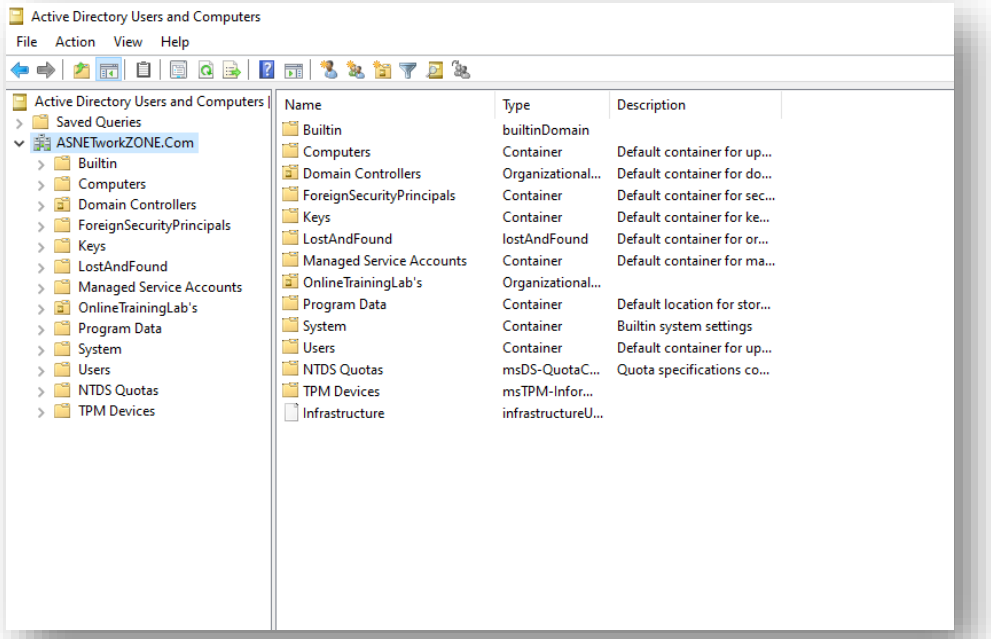
About Active Directory Domain Service (ADDS)
About ADDS database and it’s partitions
About Authentication Process in ADDS
About ADDS Object’s
Purpose of ADDS
Features of ADDS
Structure of ADDS
About FSMO Roles in ADDS
About ADDS Schema Version
Requirement of Installation ADDS





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ABOUT ACTIVE DIRECTORY DOMAIN SERVICE (ADDS)



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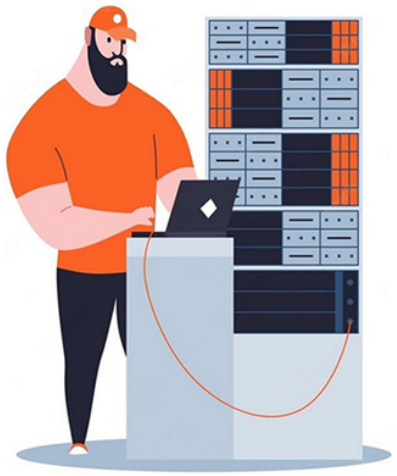
ACTIVE DIRECTORY DOMAIN SERVICE (ADDS)



Active Directory Domain Services (AD DS) is a directory service developed by Microsoft that is used to manage and organize IT resources in a networked environment, typically within a corporate or organizational setting. It serves as a centralized database to store information about users, computers, devices, and other resources in a network. AD plays a key role in ensuring the security, management, and accessibility of networked resources. Active Directory was introduced by **Microsoft Corp. in 2000**. Active Directory use KCC (Knowledge Consistency Checker) service for replication between multiple domain controllers. Active Directory is also known as hierarchical directory database.

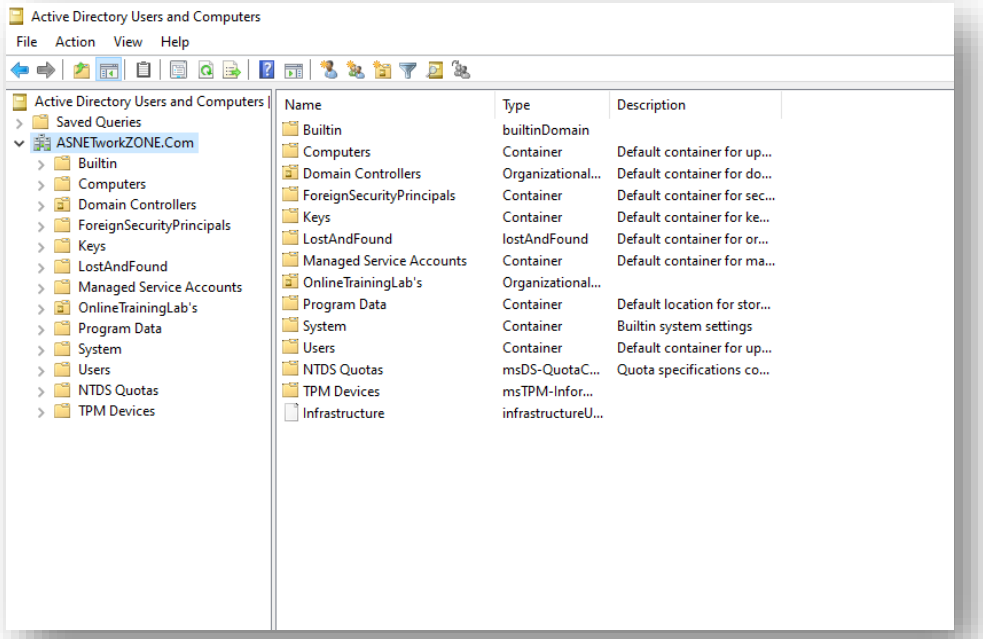
Key functions and components of Active Directory include:

- **User and Group Management:**
  AD allows administrators to create, manage, and store user accounts, groups, and other objects (like printers, computers, etc.).
- **Authentication and Authorization:**
  Authentication: AD verifies the identity of users and computers by using credentials like usernames and passwords.
  Authorization: After authentication, AD determines what resources the authenticated users can access based on group membership and permissions.
- **Centralized Management:**
  AD allows administrators to manage network resources from a central location (typically through a tool like Active Directory Users and Computers).
- **Domain Services:**
  AD operates based on the concept of domains, where a domain is a logical grouping of resources.
  Domain Controllers (DCs) are servers that store a copy of the AD database and are responsible for authenticating and authorizing users within a domain.
- **Active Directory Trusts:**
  Trust relationships allow users in one domain to access resources in another domain, enabling collaboration across different parts of an organization or between multiple organizations.
- **Group Policy:**
  Through Group Policy Objects (GPOs), administrators can enforce security settings, software installations, and other configurations across all computers in the domain.
- **DNS Integration:**
  AD integrates with DNS (Domain Name System), as DNS is used to locate domain controllers and manage network resources.
- **ADDS Replication:**
  AD data is automatically replicated across multiple domain controllers to ensure redundancy and consistency.

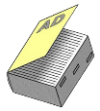


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ADDS DATABASE & IT's PARTITIONS



ADDS DATABSE & PARTITION's



Active Directory database stores in a file that is known as **NTDS.DIT**. NTDS.DIT stands for **N**ew **T**echnology **D**irectory **S**ervice.**D**irectory **I**nformation **T**ree. This file is divided into 4 logical partitions.

Schema Partition:

This logical partition is template for Active Directory database. This partition display database in form format.

It is extensive, dynamic and there is only one schema for AD in a forest.

Configuration Partition:

This logical partition contains the information about Domain, Sites & Sites Link, Domain Controller Locations and Subnets.

Domain Partition:

This logical partition contains the AD database. This partition information never replicates with other domains.

Application Partition

This partition was introduced with server 2003. This partition can be added or removed. This partition is useful when you are using AD integrated services such as DNS.

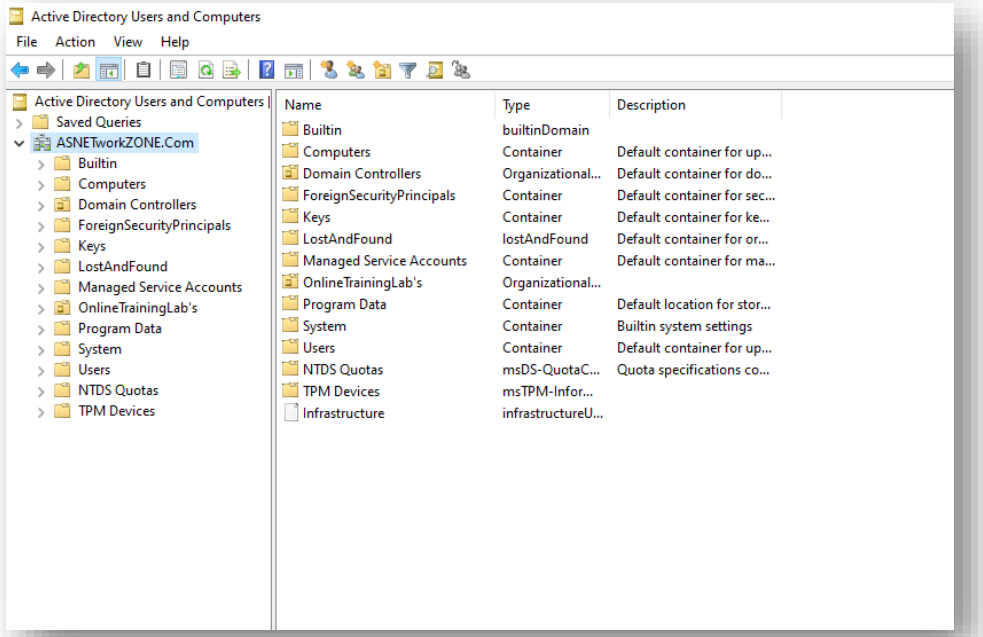
ADDS Installation Default Path

- C:\Windows\NTDS
  - Database – NTDS.DIT - ADSI.Edit
- C:\Windows\NTDS\
  - AD Logs
- C:\Windows\Sysvol
  - Scripts
  - Policy



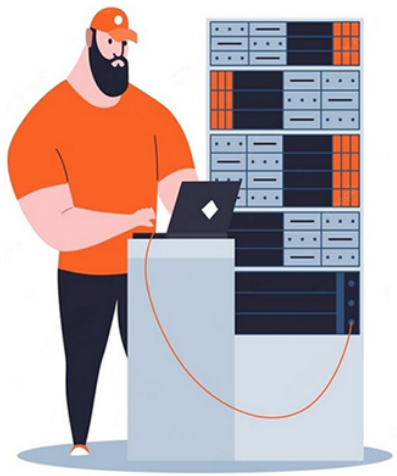
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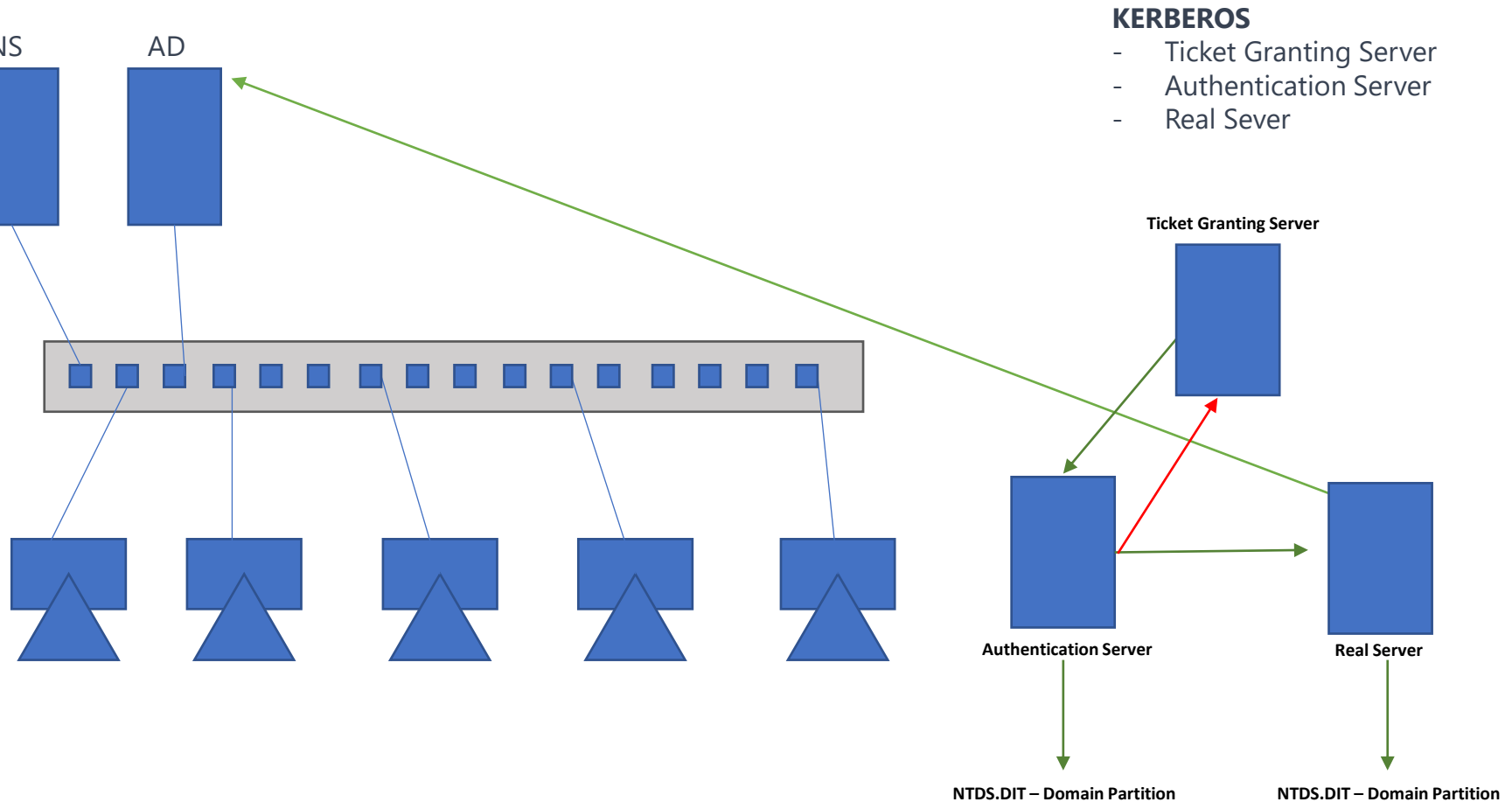
AUTHENTICATION PROCESS IN ADDS



AUTHENTICATION PROCESS IN ADDS

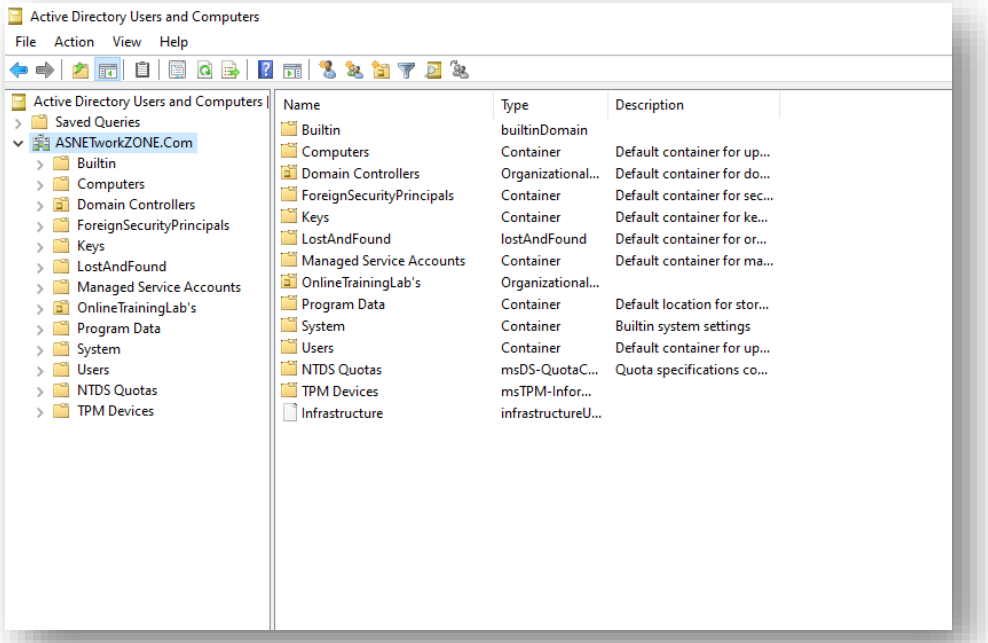
Active Directory Domain Services (AD DS) utilizes the **Kerberos protocol** as its primary authentication mechanism. Kerberos is a secure, fast, and scalable protocol designed for authenticating service requests between trusted entities on a network. Below is an overview of how the Kerberos protocol works within AD DS:





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ADDS OBJECTS



ADDS OBJECTS



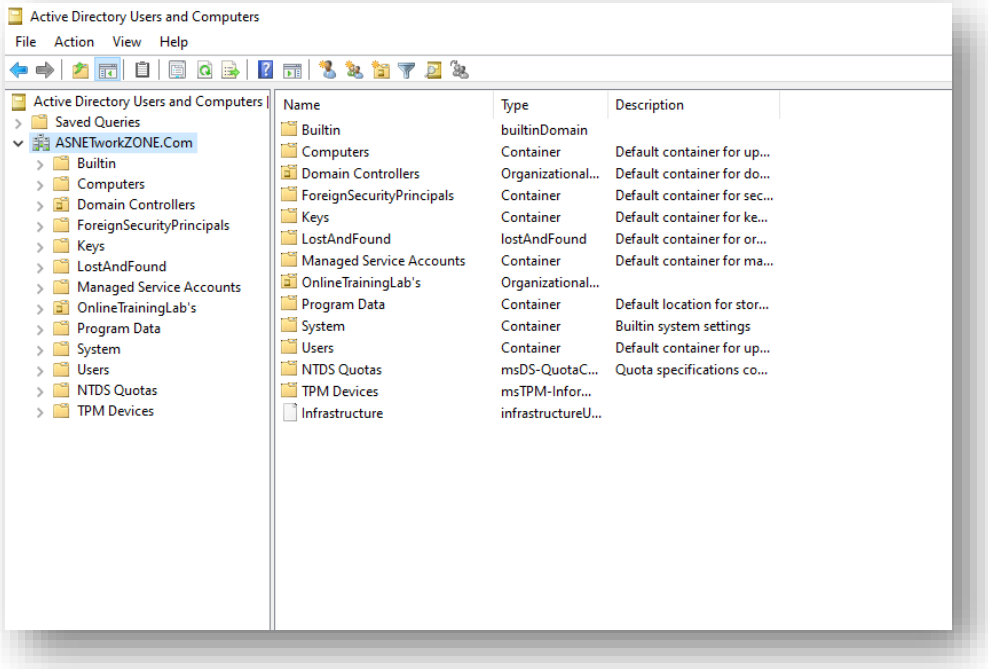
In **Active Directory Domain Services (AD DS)**), an object is any identifiable item that is stored within the directory. These objects can represent a variety of resources on the network, such as users, computers, printers, or groups. Each object in Active Directory is defined by a set of attributes, and these objects can be organized and managed through different structures like domains, organizational units (OUs), and trees.

- Users
- Groups
- Computers
- Domain Controller
- Organization Unit
- Share Folder
- Printer
- Sites
- Group Policy



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ADDS PURPOSE



ADDS PURPOSE



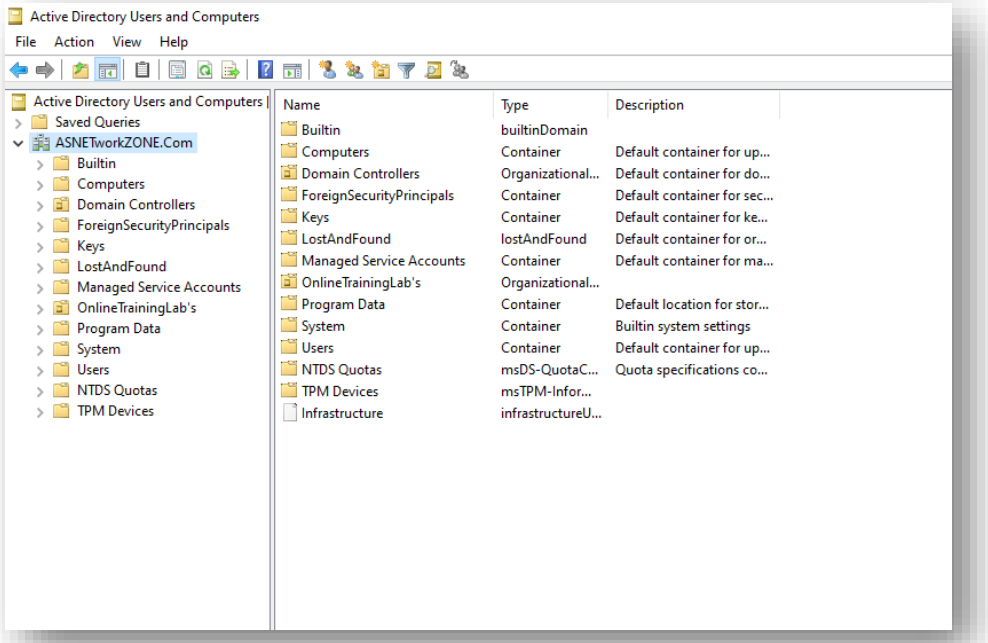
The purpose of **Active Directory Domain Services (AD DS)** is to provide a centralized directory service for managing and organizing network resources in a Windows-based environment. It enables efficient administration of users, computers, groups, and other network resources, as well as enforcing security policies and access controls. Here are the key purposes of Active Directory:

- Centralized Authentication and Authorization - typically via **Kerberos**
- User and Resource Management
- Security and Access Control
- Scalability and Delegation
- Directory-Based Service for Network Resources
- Trust Relationships and Federation
- Replication for High Availability
- Support for Cloud and Hybrid Environments
- Directory Services for Applications



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ADDS FEATURE



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ADDS FEATURE



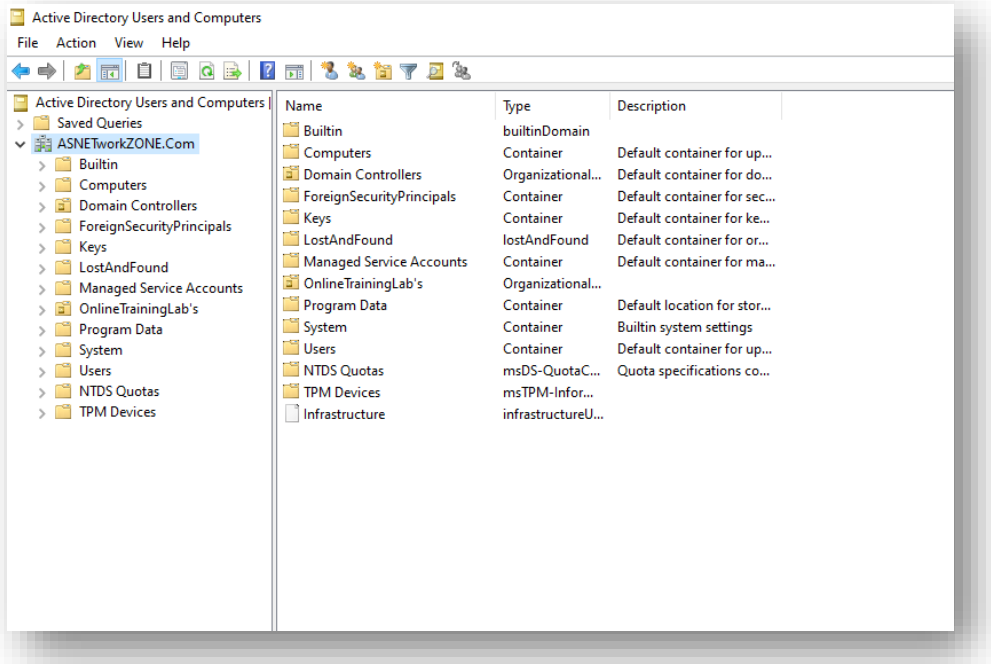
Active Directory Domain Services (AD DS) offers a wide range of features that support centralized management, security, and organization of network resources. These features help administrators manage users, computers, devices, security policies, and access control efficiently. Below are the key features of Active Directory:

- Easy administration using group policy
- Renaming domain controller name or domain name.
- Fully integrated security system with the help of Kerberos.
- Flexible to install or un-install Active Directory.
- Cross forest trust relationship.
- Site to Site replication is faster.
- Extensible (modify the schema)



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ADDS STRUCTURE



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ADDS STRUCTURE



Active Directory Domain Services (AD DS) is the core service of Active Directory and forms the backbone of directory-based identity management for users, devices, and resources. AD DS structures the directory in a hierarchical manner, allowing for organized management, scalability, and control over a network's resources. Below is a breakdown of the main components and structure of Active Directory Domain Services (AD DS):

Logical Structure

Domain - A domain is a logical grouping of network resources within a forest.

Tree - A Tree is a collection of domains that share a common namespace (e.g., company.com, hr.company.com, it.company.com).

Forest - A forest is the highest-level container in Active Directory and represents the overall directory structure. A forest can contain one or more domains

Organization Unit - An Organizational Unit (OU) is a container within a domain that can hold users, groups, computers, and other OUs.

Physical Structure

Site - A site in Active Directory represents a physical location that contains one or more subnets. Sites are used to control network traffic and replication within AD, especially in geographically dispersed networks.

Domain Controller - A Domain Controller (DC) is a server that holds a copy of the Active Directory database and is responsible for handling requests related to authentication, authorization, and directory queries.



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ABOUT FSMO ROLES IN ADDS



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FSMO ROLES IN ADDS



FSMO (Flexible Single Master Operations) roles in Active Directory Domain Services **(AD DS)** are specialized domain controller roles that ensure certain critical operations are not handled by multiple domain controllers at the same time. These roles are assigned to specific domain controllers within an Active Directory forest. There are five FSMO roles, split between two categories: forest-wide and domain-wide.

Forest Wide Role - These roles apply across the entire Active Directory forest and ensure consistency in forest-wide operations.

Schema Master Role

Purpose: The Schema Master role controls all updates and modifications to the Active Directory schema. This includes adding, modifying, or deleting classes and attributes in the schema.
Assigned to: Only one domain controller in the entire forest holds this role.
Critical Operations: Changes to the AD schema (e.g., adding new attributes, classes) can only be made by the domain controller holding this role.

Domain Naming Master

Purpose: The Domain Naming Master controls the addition and removal of domains in the forest. It ensures that domain names are unique and can be properly referenced in the forest.
Assigned to: Only one domain controller in the entire forest holds this role.
Critical Operations: Adding/removing domains or renaming the domain in the forest.



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FSMO ROLES IN ADDS



FSMO (Flexible Single Master Operations) roles in Active Directory Domain Services **(AD DS)** are specialized domain controller roles that ensure certain critical operations are not handled by multiple domain controllers at the same time. These roles are assigned to specific domain controllers within an Active Directory forest. There are five FSMO roles, split between two categories: forest-wide and domain-wide.

Domain Wide Role - These roles are used within a single domain in the forest. The roles impact domain-specific operations.

RID Master Role

Purpose: The RID Master assigns unique Relative Identifiers (RIDs) to domain controllers. These RIDs are used in security identifiers (SIDs) for users, groups, and computers.
Assigned to: Only one domain controller per domain holds this role.
Critical Operations: Ensures that each domain controller can issue unique RIDs for objects in the domain to avoid duplication of SIDs.

PDC Emulator

Purpose: The PDC Emulator role provides backward compatibility with older Windows NT-based clients and ensures time synchronization within the domain. It also handles password changes, lockouts, and group policy processing.
Assigned to: Only one domain controller per domain holds this role.

- Critical Operations: Handles time synchronization for all domain controllers.
- Acts as a password change master and helps in handling lockout and replication of password changes.
- It processes Group Policy updates.
- Handles authentication requests for Windows NT 4.0 clients.



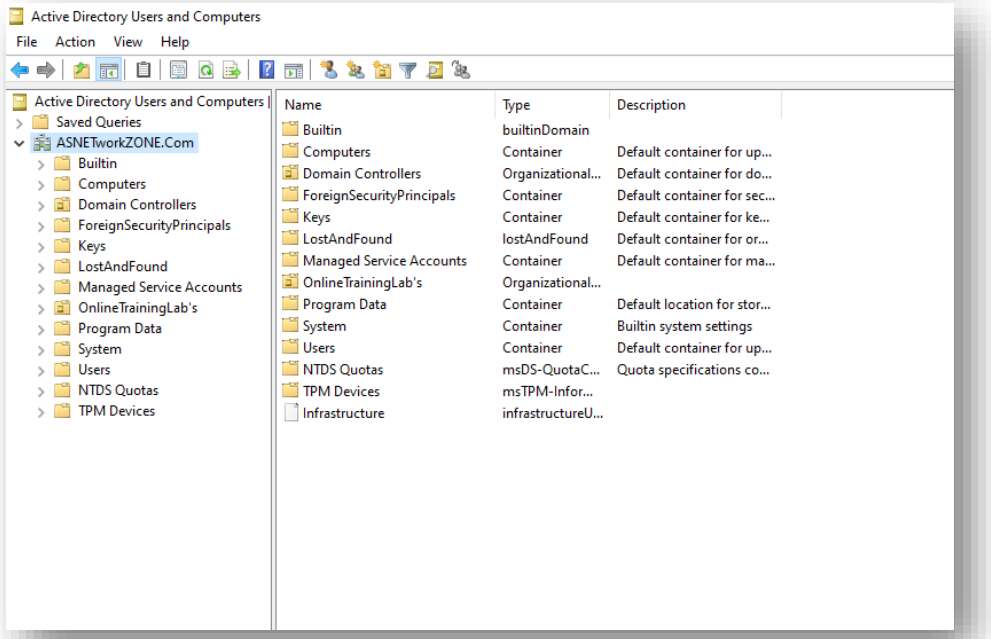
Infrastructure Master Role

Purpose: The Infrastructure Master role is responsible for updating references to objects in other domains. For example, if a user from another domain is referenced in a group in the current domain, the Infrastructure Master ensures that the user object is up to date.
Assigned to: Only one domain controller per domain holds this role.
Critical Operations: Updates references to objects from other domains (especially in multi-domain forests). This is important when there are domain trusts or cross-domain object references.

NOTE: In case of failure or the need to move a role, administrators can use tools like ntdsutil or Active Directory Users and Computers to transfer or seize the roles.

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ADDS SCHEMA VERSION



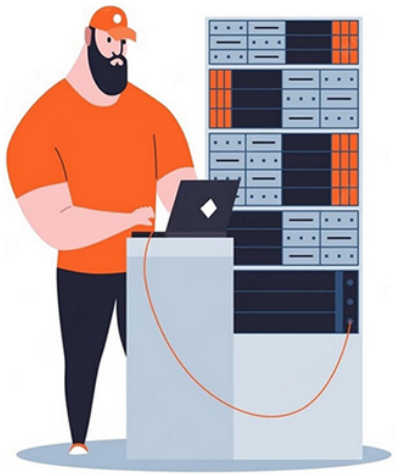
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FSMO ROLES IN ADDS



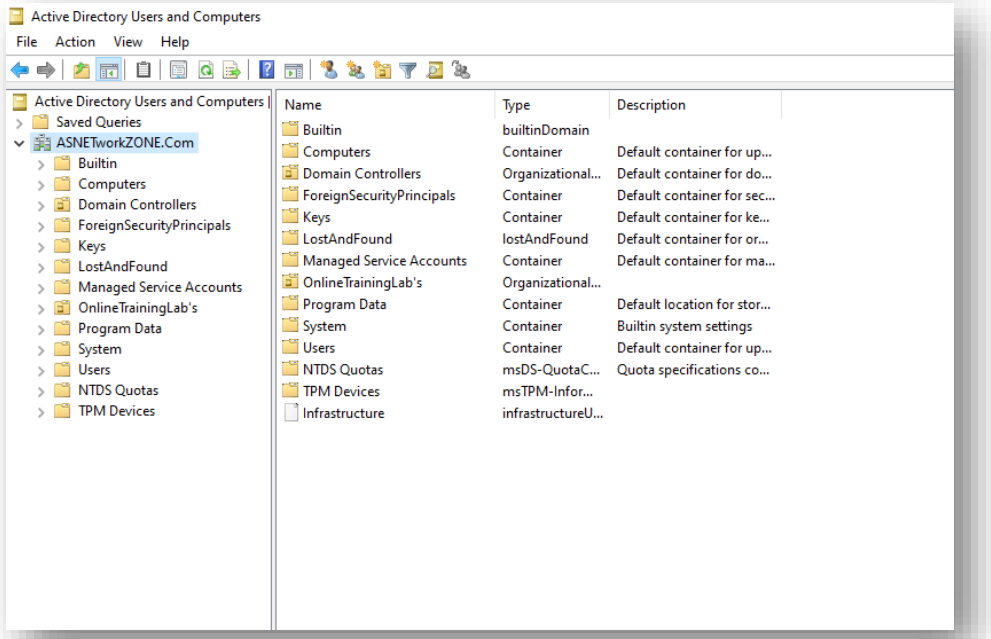
The **Active Directory Domain Services (AD DS)** Schema Version refers to the version of the schema that is used within an Active Directory environment. The schema defines the types of objects and attributes that can be stored in Active Directory, and the schema version is tied to the version of Windows Server that is running in the AD DS forest.

Version	Serve Operating System
91	Windows Server 2025
88	Windows Server 2022
88	Windows Server 2019
87	Windows Server 2016
69	Windows Server 2012 R2
56	Windows Server 2012
47	Windows Server 2008 R2
44	Windows Server 2008 RTM
31	Windows Server 2003 R2
30	Windows Server 2003 RTM, Windows 2003 Service Pack 1, Windows 2003 Service Pack 2
13	Windows Server 2000



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REQUIREMENT OF INSTALLATION ADDS



REQUIREMENT OF INSTALLATION ADDS



- Windows Server O/S
- Admin rights on server
- Static IP Address [192.168.0.111/24]
- Unique Domain Name
- DNS Server
- Directory Service Restore Mode Password
- Forest Functional Level
- Domain Functional Level
- Minimum 250 MB Disk Space
    - 200 MB – Database [NTDS.DIT]
    - 50MB – Log Files



